**Name:** Jay Arre Talosig
**Subject & Section:** Reinforcement Learning - COM231ML
**Professor:** Florencio V. Contreras Jr.

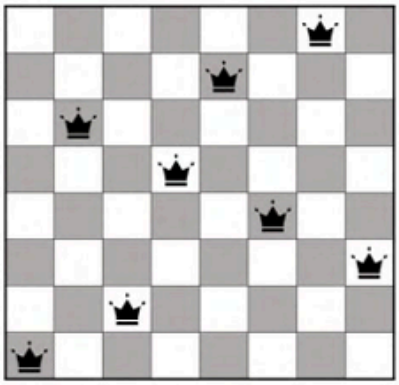
## Performance Task Problem Representation in REINFORCEMENT LEARNING

**Objective(s):**

*At the end of the exercise, the students should be able to:*

- Examine the correct representation of Reinforcement Learning-related problems through appropriate characterization; and
- Determine the distinctive properties of the methods of problem representation.

**Instructions:**



**The Eight-Queens Problem:** Place eight (8) queens on a chessboard so that no queen attacks another. A queen may attack any piece in the same row, column, or diagonal.

1. Define the following problem attributes based on your perspective: **(5 items x 3 points)**
   a. The agent(s)
   b. The state space description
   c. An initial state
   d. A goal state
   e. The possible action

2. Analyze the position of the queens on the chessboard at the left. Describe the situation or arrangement of the queens on the board based on your analysis. **(5 points)**

3. Answer the following questions based on your defined attributes and problem analysis: **(4 items x 5 points)**
   a. In your perspective, what problem representation method would best fit the eight-queen problem and why?
   b. If you are to write a detailed solution, including the set of production rules for the problem, what would be your initial approach? Elaborate on your answer.
   c. Would you recommend the implementation of the problem reduction approach in this problem? Why or why not?
   d. Based on your defined attributes, can the problem be broken down into smaller sub-problems that would have a good impact on its action cost function? Why or why not?

4. Illustrate a goal state for the given problem. **(10 points)**

**Grading rubric for the essay:**

| Criteria | Performance Indicator | Points |
|---|---|---|
| Content | Correct ideas, concepts, and/or examples were included. | 20 |
| Organization | The presentation of ideas was generally organized. | 30 |
| | **TOTAL** | **50** |

**Grading rubric for the illustration:**

| Criteria | Performance Indicator | Points |
|---|---|---|
| Content | Accurate arrangement of the chess queens was illustrated. | 20 |
| Organization | The design was generally organized. | 20 |
| | **TOTAL** | **40** |

***Note:*** *DO NOT COPY AND PASTE. Students who copy and paste their work from any website or their classmates will automatically receive a failing mark for this assessment task.*

**1. Problem Attributes**

• a. Agent(s): A single problem-solving agent (the “queen-placer”) that perceives the current board state and decides where to place the next queen. It’s modeled as one agent making sequential decisions rather than 8 separate agents, since every placement affects shared constraints across the whole board.

• b. State space: All possible partial/complete arrangements of queens on the 8×8 grid. A compact representation: an array of length 8 where the index = column and the value = the row occupied in that column. Full space ≈ $8^8$ raw assignments, though constraint checking prunes this drastically.

• c. Initial state: An empty board that has zero queens placed. For local-search variants, we could instead start with all 8 queens placed randomly, one per column, containing conflicts.

• d. Goal state: Any complete configuration of 8 queens where no two share a row, column, or diagonal.

• e. Possible actions: Place a queen in an unattacked square of the next open column (incremental construction); or, in local-search formulations, move a queen to a different row within its own column to reduce conflicts.

**2. Board Analysis**

The queens in the image form a stair-step pattern descending from the upper-right toward the lower-left with each queen sitting in a different row and column, satisfying those two constraints. But several of them line up along the same diagonal, meaning they threaten each other diagonally. So this is a non-goal conflicting state: orderly-looking, but it fails the diagonal non-attack condition that’s central to the problem.

**3. Reflection Questions**

• a. Best representation method: State-space search, implemented as a rule-based (production) system, fits best. Each queen placement transforms one state into another, goal-testing happens after each step, and it pairs naturally with backtracking with no need to model complex nested subgoals.

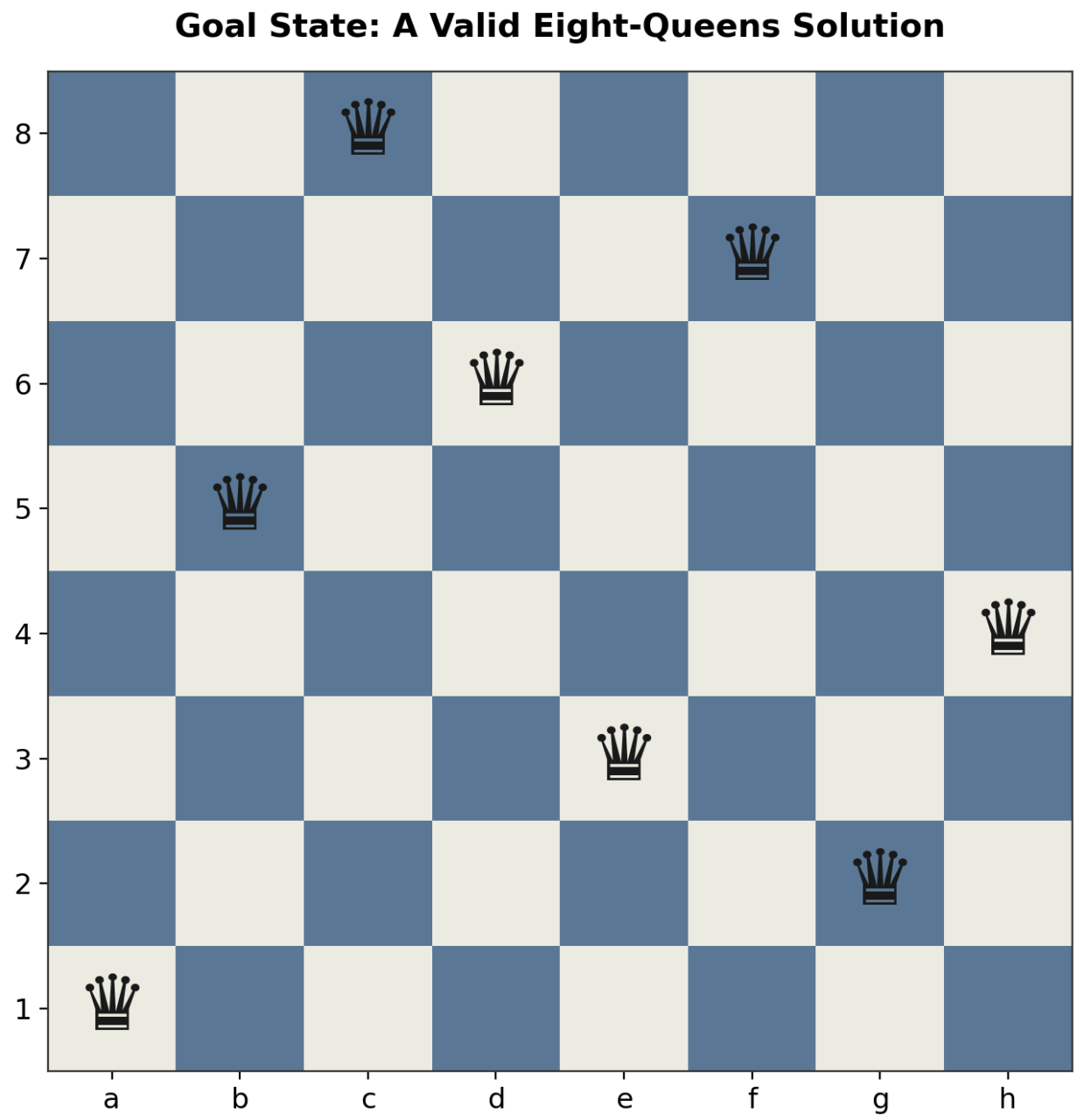
• b. Production rules / initial approach: Something like: IF placing a queen in column c, row r conflicts with no previously placed queen (same row, or |row diff| = |col diff| for diagonals) THEN place it and move to column c+1; ELSE try the next row; IF all rows in a column are exhausted THEN backtrack to the previous column. This incremental-and-backtrack strategy checks one column at a time.

• c. Problem reduction: Not really recommended. Problem reduction (AND-OR graphs) shines when a problem splits into independent subproblems solved separately then merged (like Tower of Hanoi). In eight-queens, every placement constrains every other one through shared rows/diagonals, so subproblems stay tightly coupled that is reduction adds overhead without real simplification.

• d. Sub-problems and action cost: Only partially useful. You could try splitting the board into regions for example left half/right half, but since diagonals and rows span the entire board, merging sub-solutions often creates new conflicts. Cost reduction comes more reliably from constraint propagation/early pruning than from structural decomposition.

**4. Goal State Illustration**

Here's a verified valid solution (columns a–h, one queen per row/column, no diagonal conflicts):




**Positions**: a1, b5, c8, d6, e3, f7, g2 ,h4